ASTANA IT UNIVERSITY

Computer networks



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Lecture 9

Application Layer

Objectives

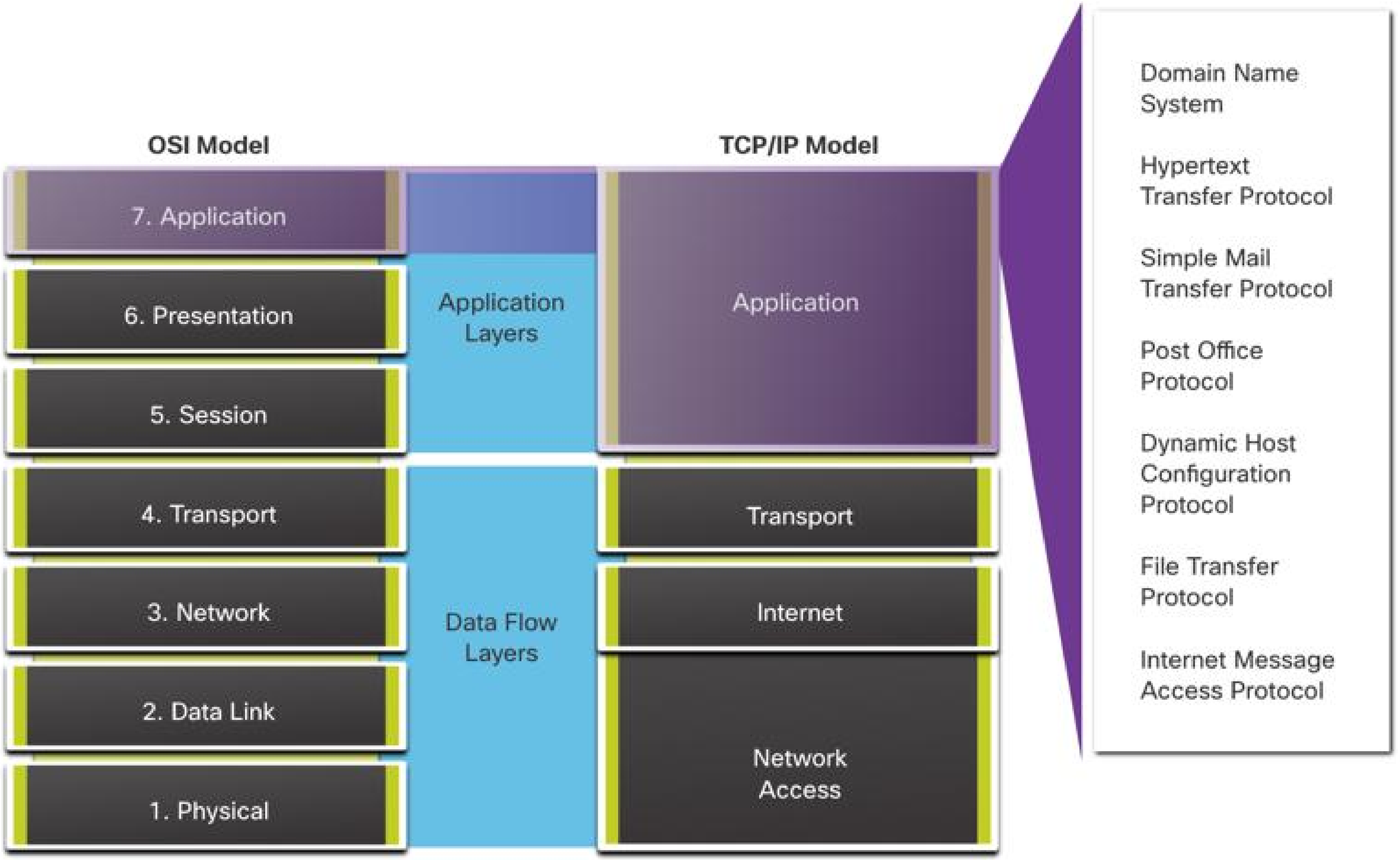
Application, Presentation, and Session	Web and Email Protocols	IP Addressing Services

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Application, Presentation, and Session

Application Layer

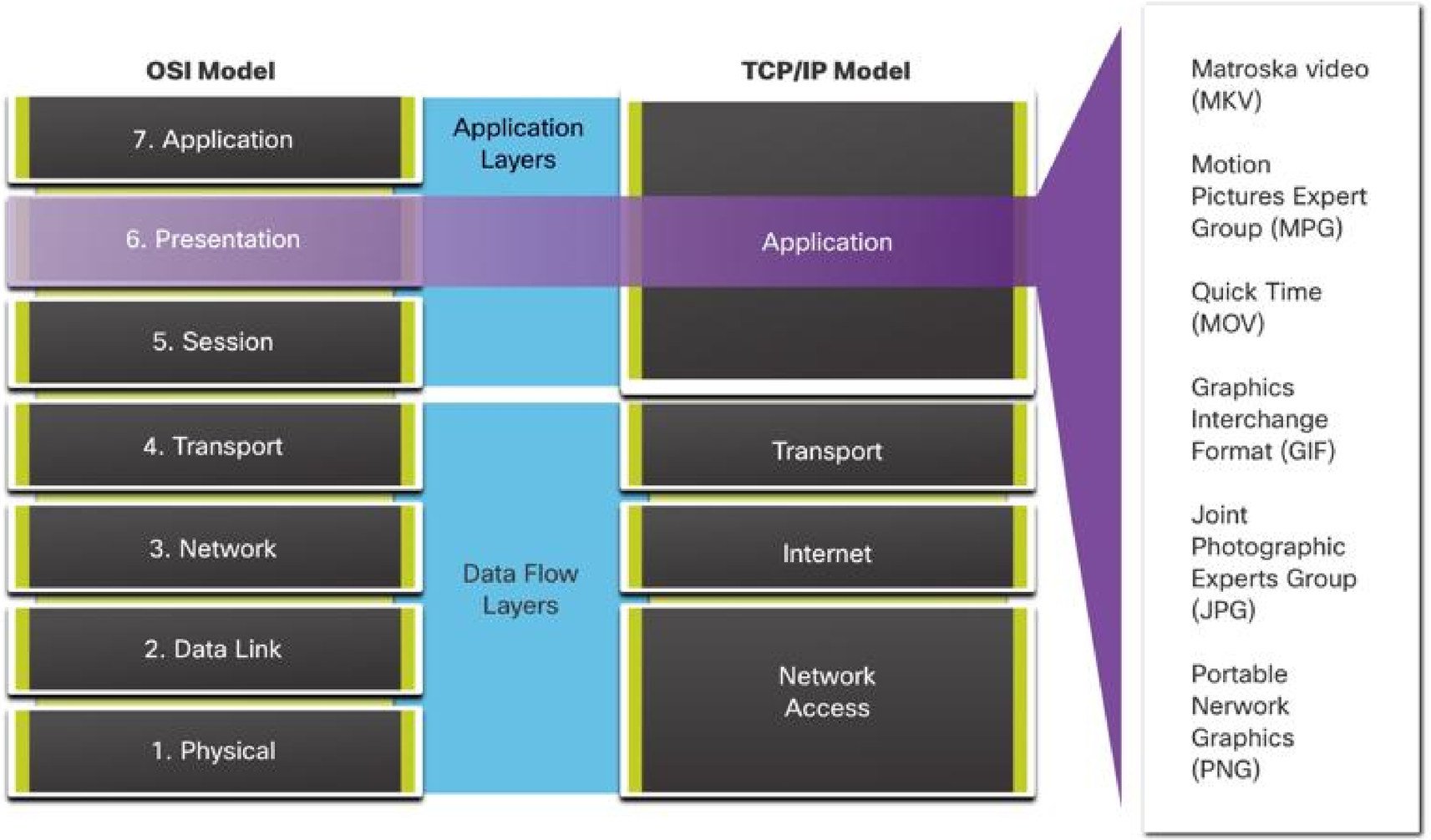
- The upper three layers of the OSI model (application, presentation, and session) define functions of the TCP/IP application layer.
- The application layer provides the interface between the applications used to communicate, and the underlying network over which messages are transmitted.
- Some of the most widely known application layer protocols include HTTP, FTP, TFTP, IMAP and DNS.


OSI Model
TCP/IP Model
7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical
Application Layers
Data Flow Layers
Application
Transport
Internet
Network Access
Domain Name System
Hypertext Transfer Protocol
Simple Mail Transfer Protocol
Post Office Protocol
Dynamic Host Configuration Protocol
File Transfer Protocol
Internet Message Access Protocol

Presentation Layer

The presentation layer has three primary functions:

- Formatting, or presenting, data at the source device into a compatible format for receipt by the destination device
- Compressing data in a way that can be decompressed by the destination device
- Encrypting data for transmission and decrypting data upon receipt



Session Layer

The session layer functions:

- It creates and maintains dialogs between source and destination applications.
- It handles the exchange of information to initiate dialogs, keep them active, and to restart sessions that are disrupted or idle for a long period of time.

TCP/IP Application Layer Protocols

- The TCP/IP application protocols specify the format and control information necessary for many common internet communication functions.
- Application layer protocols are used by both the source and destination devices during a communication session.
- For the communications to be successful, the application layer protocols that are implemented on the source and destination host must be compatible.

TCP/IP Application Layer Protocols

Name System

DNS - Domain Name System (or Service)

- TCP, UDP client 53
- Translates domain names, such as cisco.com, into IP addresses.

Host Config

DHCP - Dynamic Host Configuration Protocol

- UDP client 68, server 67
- Dynamically assigns IP addresses to be re-used when no longer needed

Web

HTTP - Hypertext Transfer Protocol

- TCP 80, 8080
- A set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the World Wide Web

A network administrator is verifying the configuration of a newly installed host by establishing an FTP connection to a remote server. What is the highest layer of the protocol stack that the network administrator is using for this operation?

A. application
B. presentation
C. session
D. transport
E. internet
F. data link

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Web and Email Protocols

Hypertext Transfer Protocol and Hypertext Markup Language

When a web address or Uniform Resource Locator (URL) is typed into a web browser, the web browser establishes a connection to the web service. The web service is running on the server that is using the HTTP protocol.

To better understand how the web browser and web server interact, examine how a web page is opened in a browser.

Step 1

The browser interprets the three parts of the URL:

- http (the protocol or scheme)
- www.cisco.com (the server name)
- index.html (the specific filename requested)



Hypertext Transfer Protocol and Hypertext Markup Language

Step 1

The browser interprets the three parts of the URL:

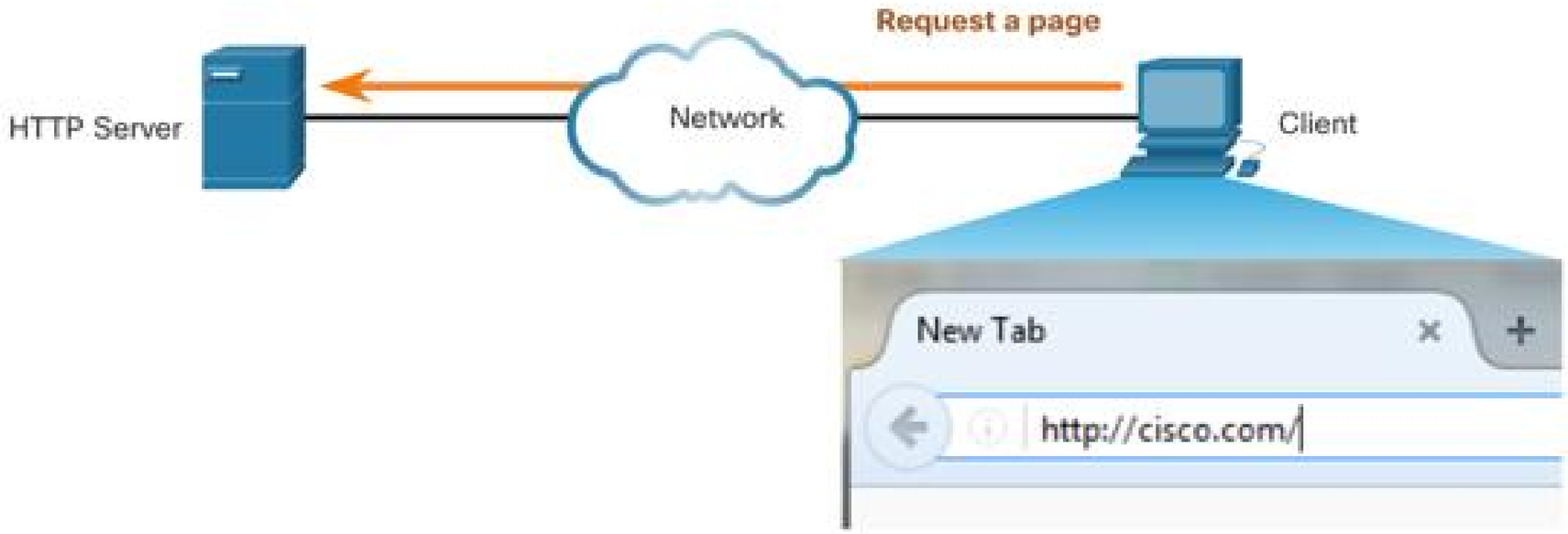
- http (the protocol or scheme)
- www.cisco.com (the server name)
- index.html (the specific filename requested)



Hypertext Transfer Protocol and Hypertext Markup Language (Cont.)

Step 2
The browser then checks with a name server to convert www.cisco.com into a numeric IP address, which it uses to connect to the server.

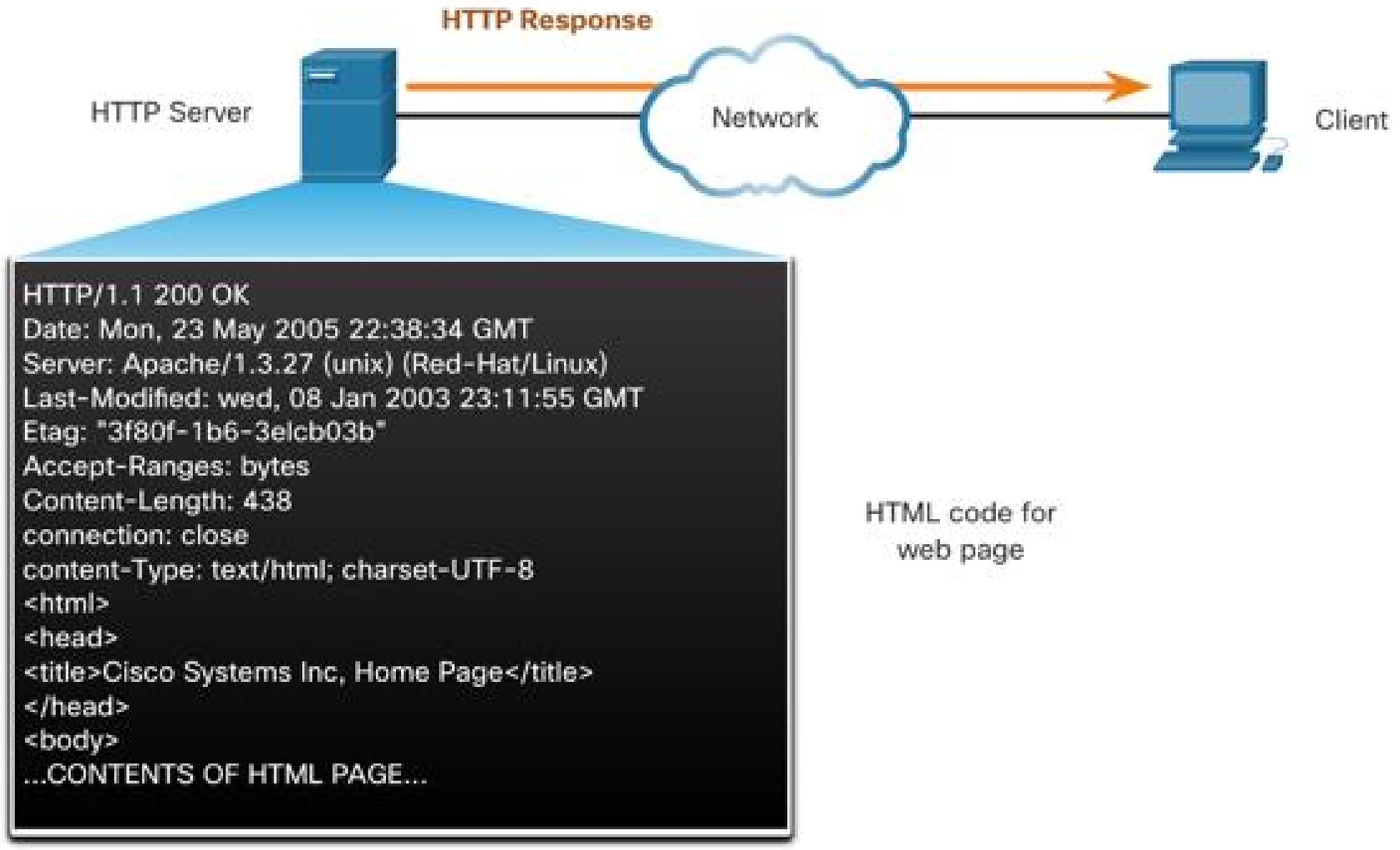
The client initiates an HTTP request to a server by sending a GET request to the server and asks for the index.html file.



Hypertext Transfer Protocol and Hypertext Markup Language (Cont.)

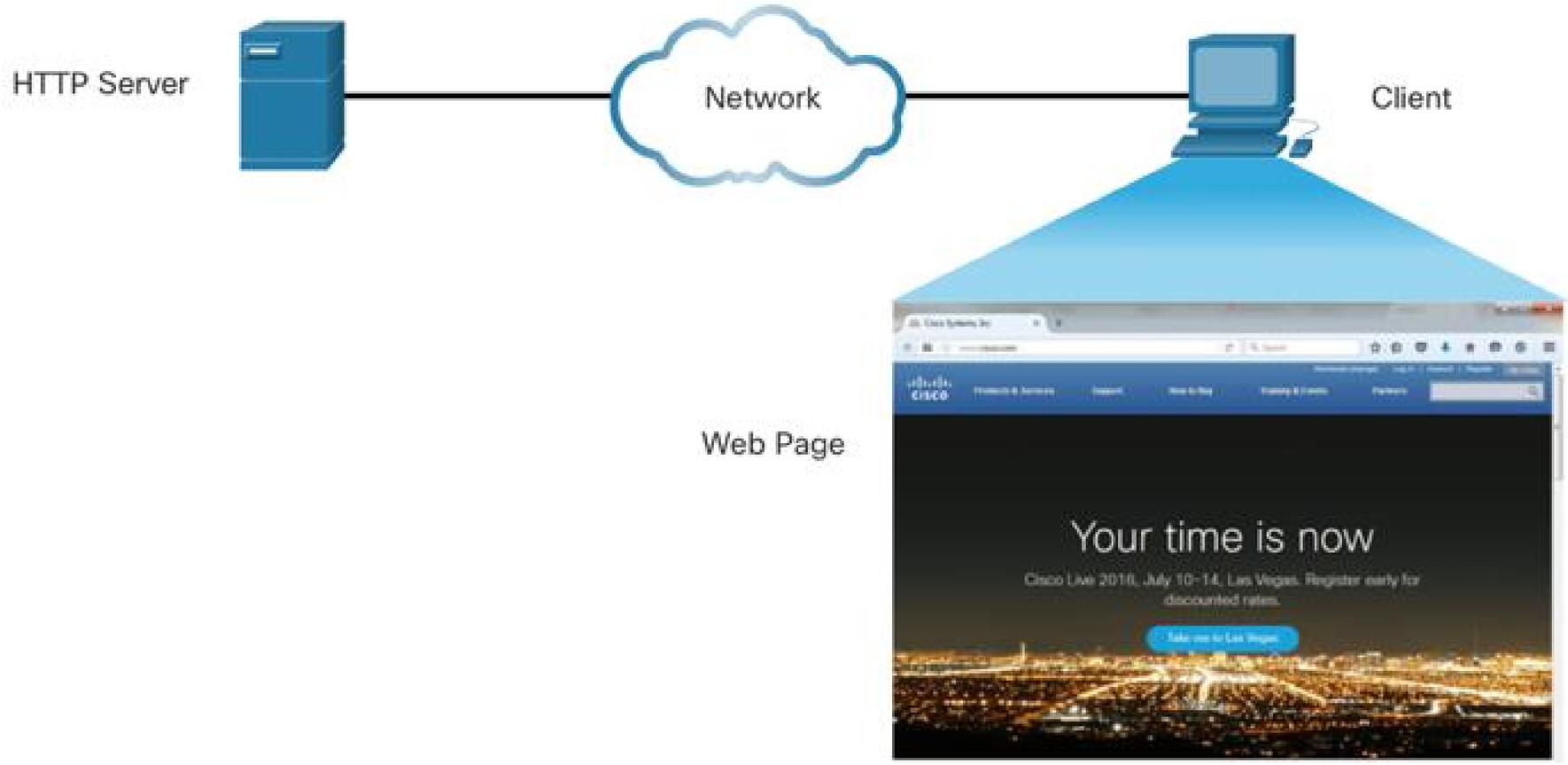
Step 3

In response to the request, the server sends the HTML code for this web page to the browser.



Hypertext Transfer Protocol and Hypertext Markup Language (Cont.)

Step 4
The browser deciphers the HTML code and formats the page for the browser window.



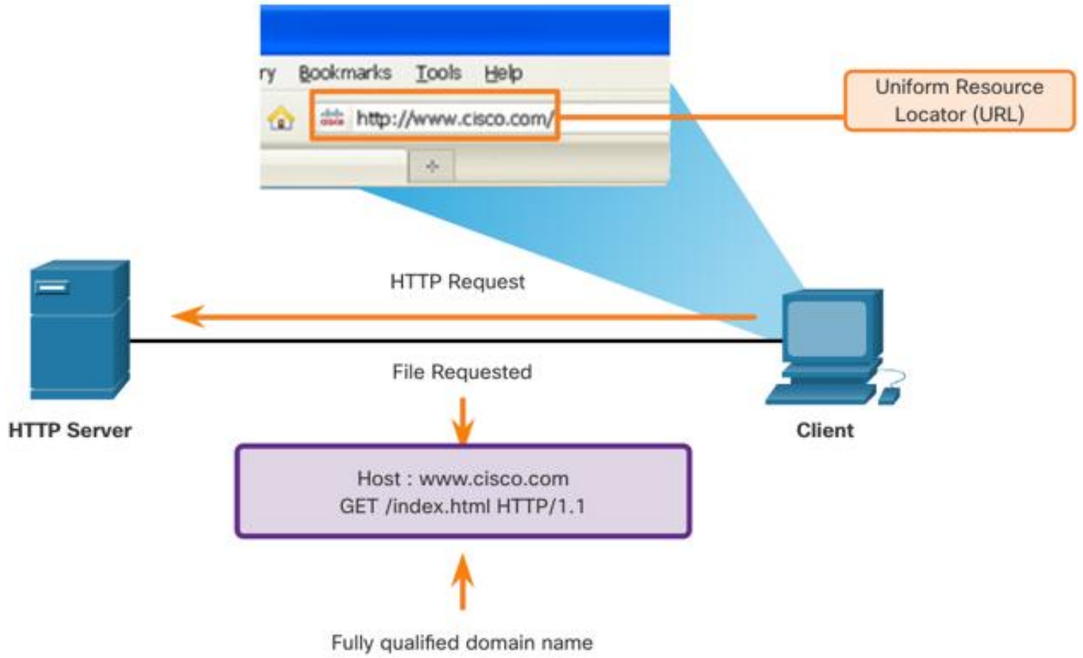
Web and Email Protocols

HTTP and HTTPS

HTTP is a request/response protocol that specifies the message types used for that communication.

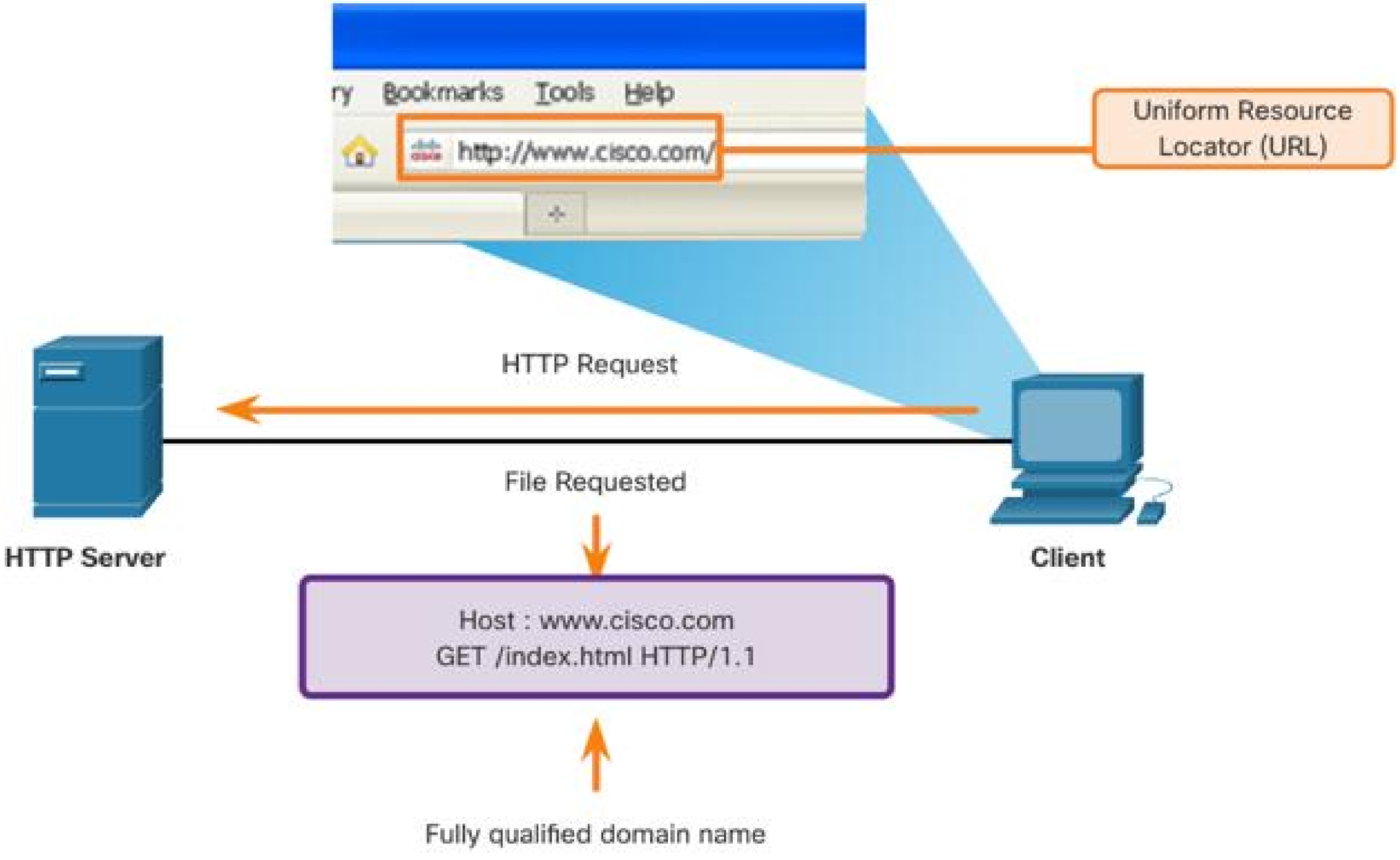
The three common message types are GET, POST, and PUT:

- **GET** - This is a client request for data. A client (web browser) sends the GET message to the web server to request HTML pages.
- **POST** - This uploads data files to the web server, such as form data.
- **PUT** - This uploads resources or content to the web server, such as an image.



Web and Email Protocols

HTTP and HTTPS

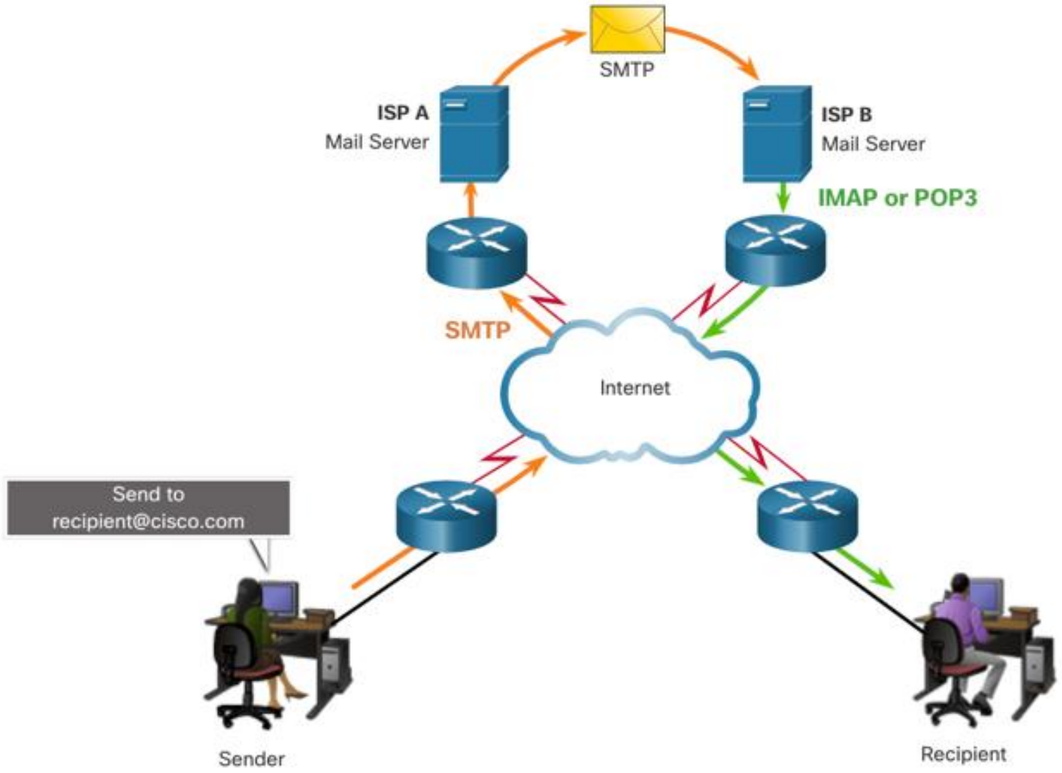


Email Protocols

Email is a store-and-forward method of sending, storing, and retrieving electronic messages across a network. Email messages are stored in databases on mail servers. Email clients communicate with mail servers to send and receive email.

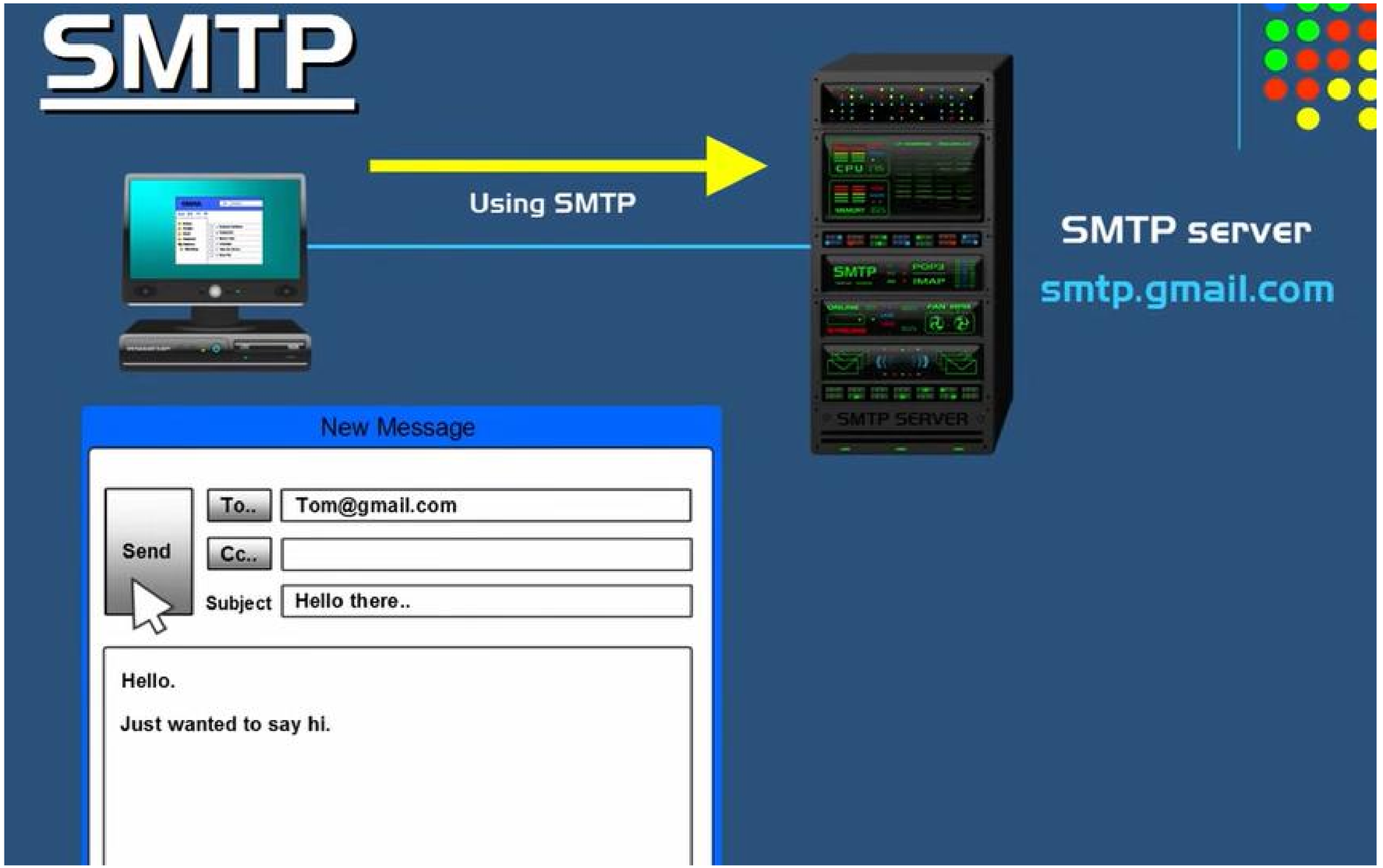
The email protocols used for operation are:

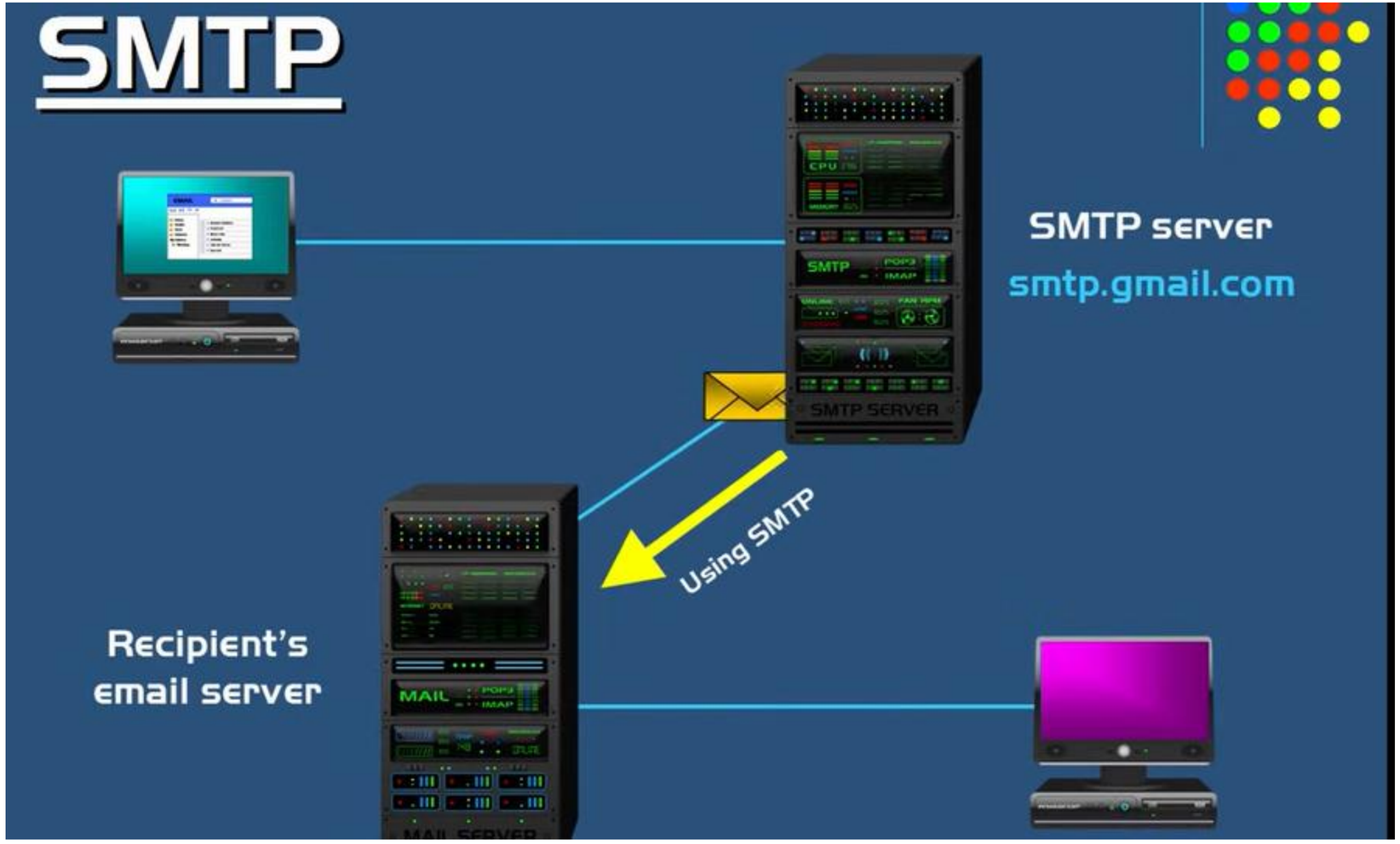
- Simple Mail Transfer Protocol (SMTP) – used to send mail.
- Post Office Protocol (POP) & IMAP – used for clients to receive mail.

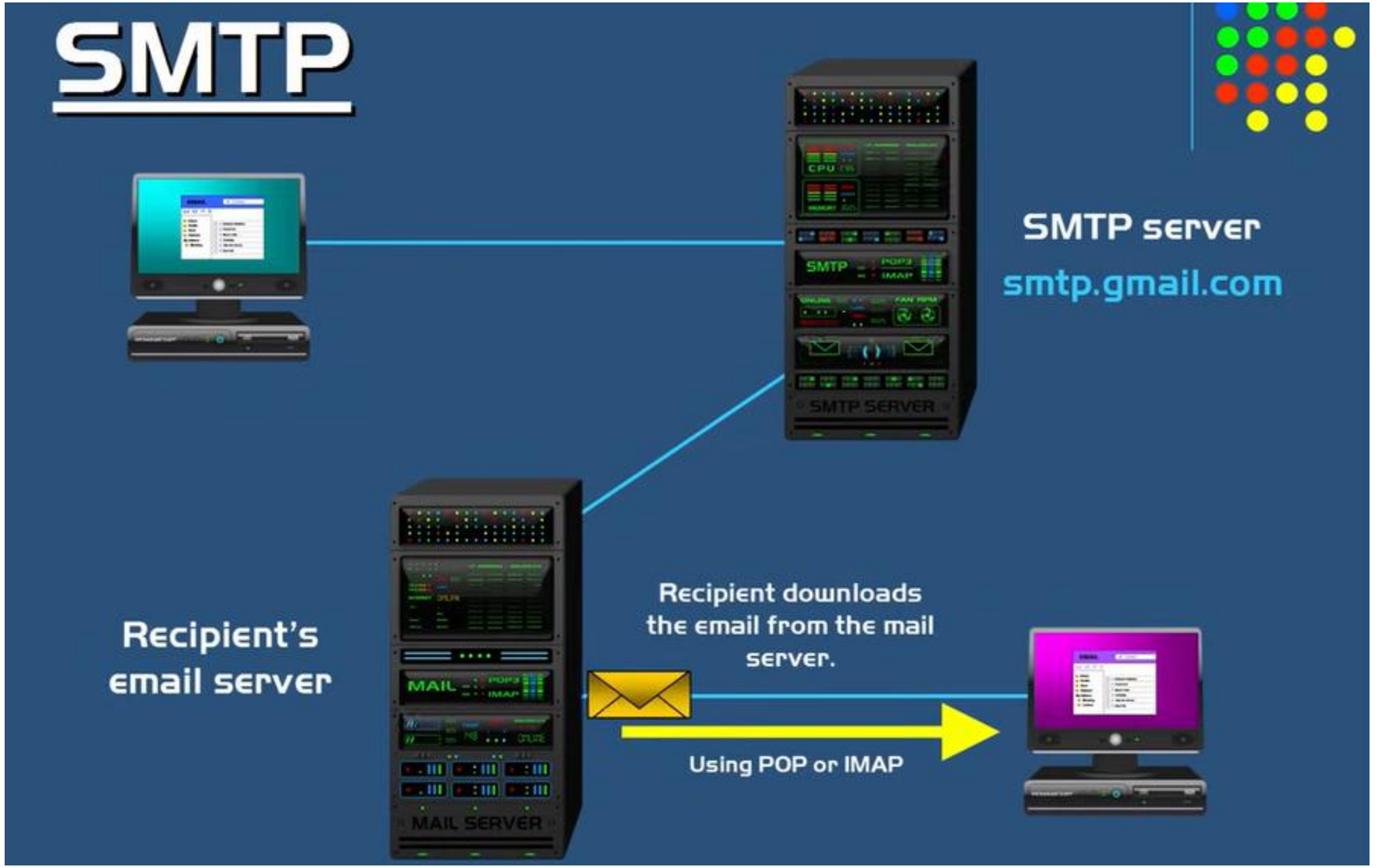


Simple Mail Transfer Protocol (SMTP)



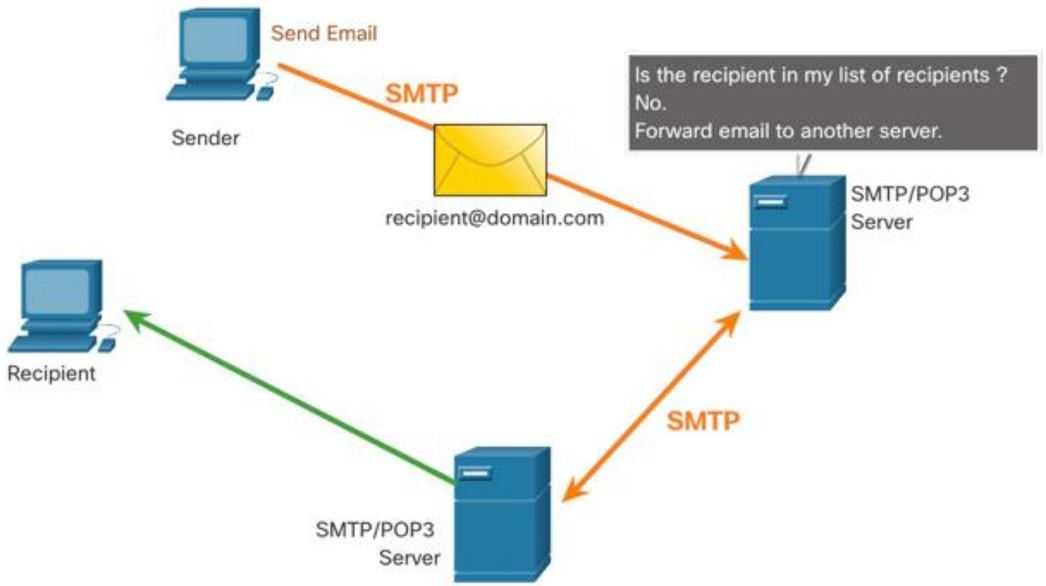

SMTP
Using SMTP
SMTP server
smtp.gmail.com
SMTP SERVER
New Message
Send
To..
Tom@gmail.com
Cc..
Subject
Hello there..
Hello.
Just wanted to say hi.


SMTP
SMTP server
smtp.gmail.com
SMTP SERVER
Using SMTP
Recipient's
email server
MAIL


SMTP
SMTP server
smtp.gmail.com
SMTP SERVER
Recipient's email server
MAIL SERVER
Recipient downloads the email from the mail server.
Using POP or IMAP

Simple Mail Transfer Protocol (SMTP)

- When a client sends email, the client SMTP process connects with a server SMTP process on well-known port 25.
- After the connection is made, the client attempts to send the email to the server across the connection.
- When the server receives the message, it either places the message in a local account, if the recipient is local, or forwards the message to another mail server for delivery.
- The destination email server may not be online or may be busy. If so, SMTP spools messages to be sent at a later time.



Note: SMTP message formats require a message header (recipient email address & sender email address) and a message body.

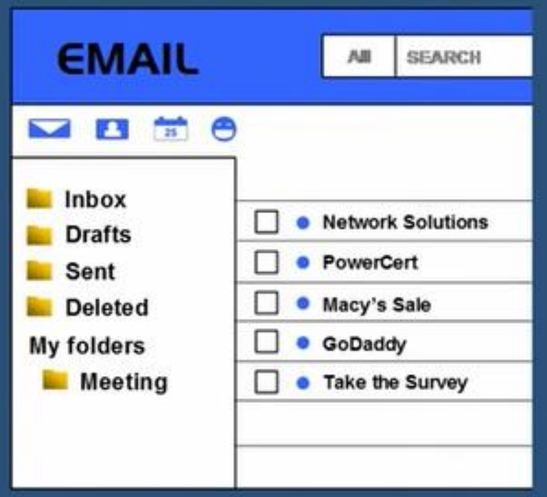
POP3

The simplest of the 2 protocols.









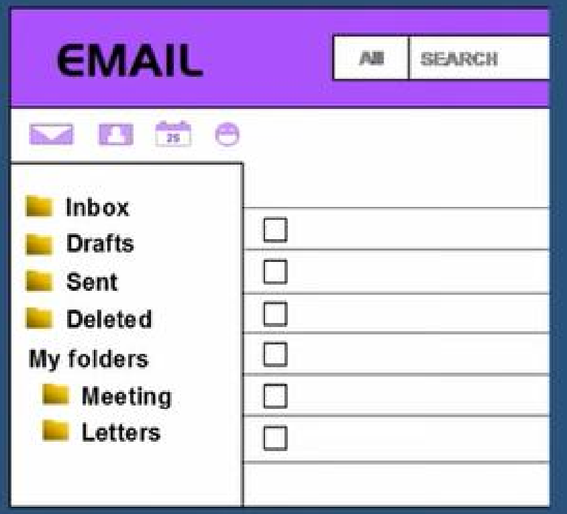
The only thing that POP3 does, is download the email to your device from a mail server.

POP3

In POP3, the email is deleted on the mail server once it's downloaded to a device.







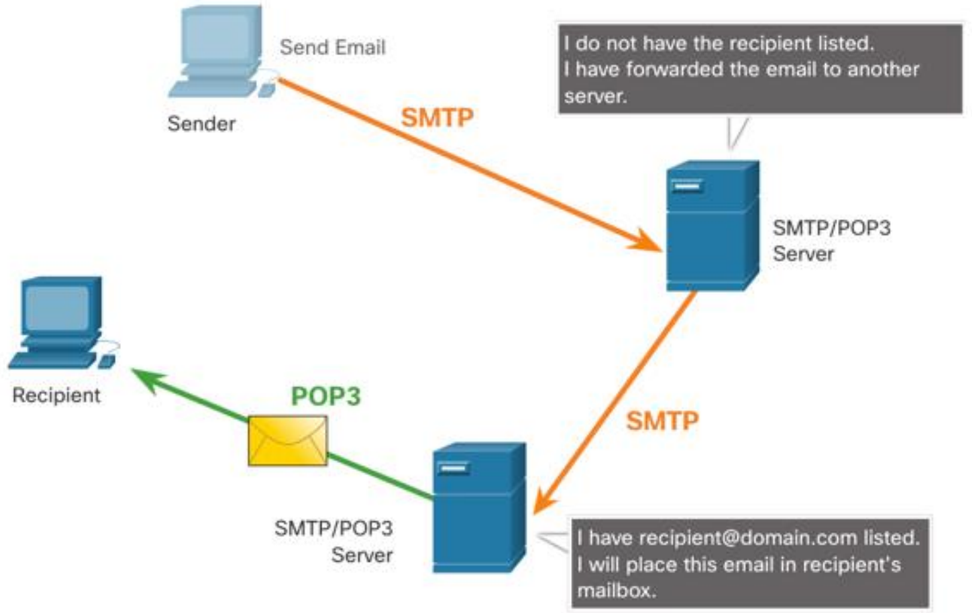




POP

POP is used by an application to retrieve mail from a mail server. When mail is downloaded from the server to the client using POP the messages are then deleted on the server.

- The server starts the POP service by passively listening on TCP port 110 for client connection requests.
- When a client wants to make use of the service, it sends a request to establish a TCP connection with the server.
- When the connection is established, the POP server sends a greeting.
- The client and POP server then exchange commands and responses until the connection is closed or aborted.



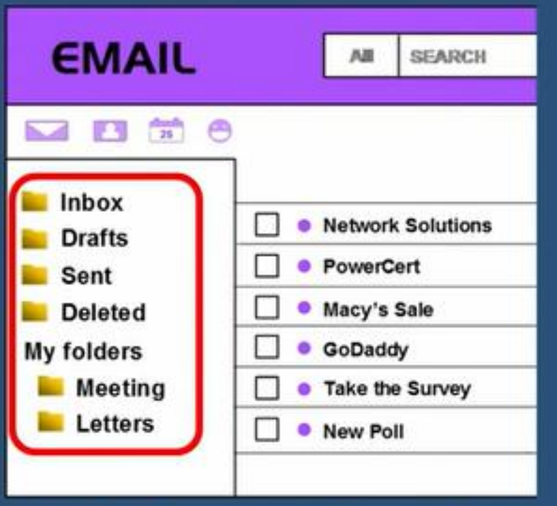
Note: Since POP does not store messages, it is not recommended for small businesses that need a centralized backup solution.

IMAP

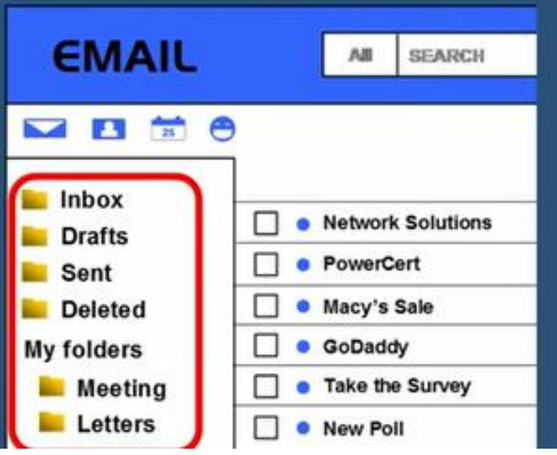
IMAP allows you to view your email, that's on the server, from multiple devices.







IMAP does synchronize.



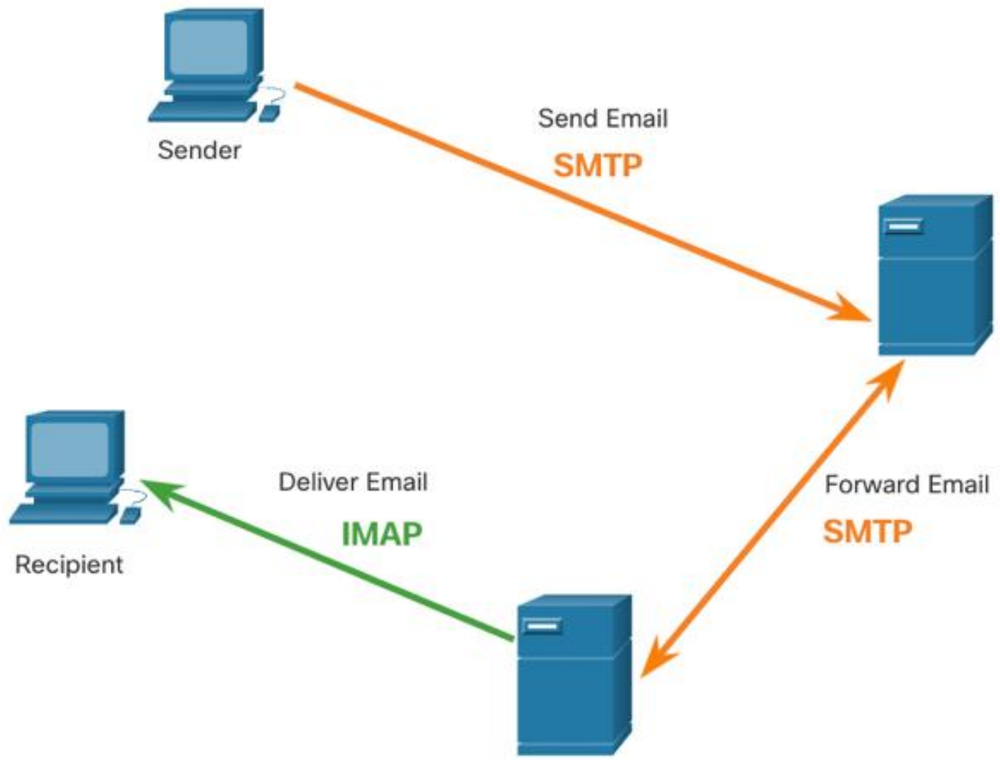
IMAP caches local copies of the email onto all of your devices.



IMAP

IMAP is another protocol that describes a method to retrieve email messages.

- Unlike POP, when a user connects to an IMAP server, copies of the messages are downloaded to the client application. The original messages are kept on the server until manually deleted.
- When a user decides to delete a message, the server synchronizes that action and deletes the message from the server.



IP Addressing Services

IP Addressing Services

Domain Name Service

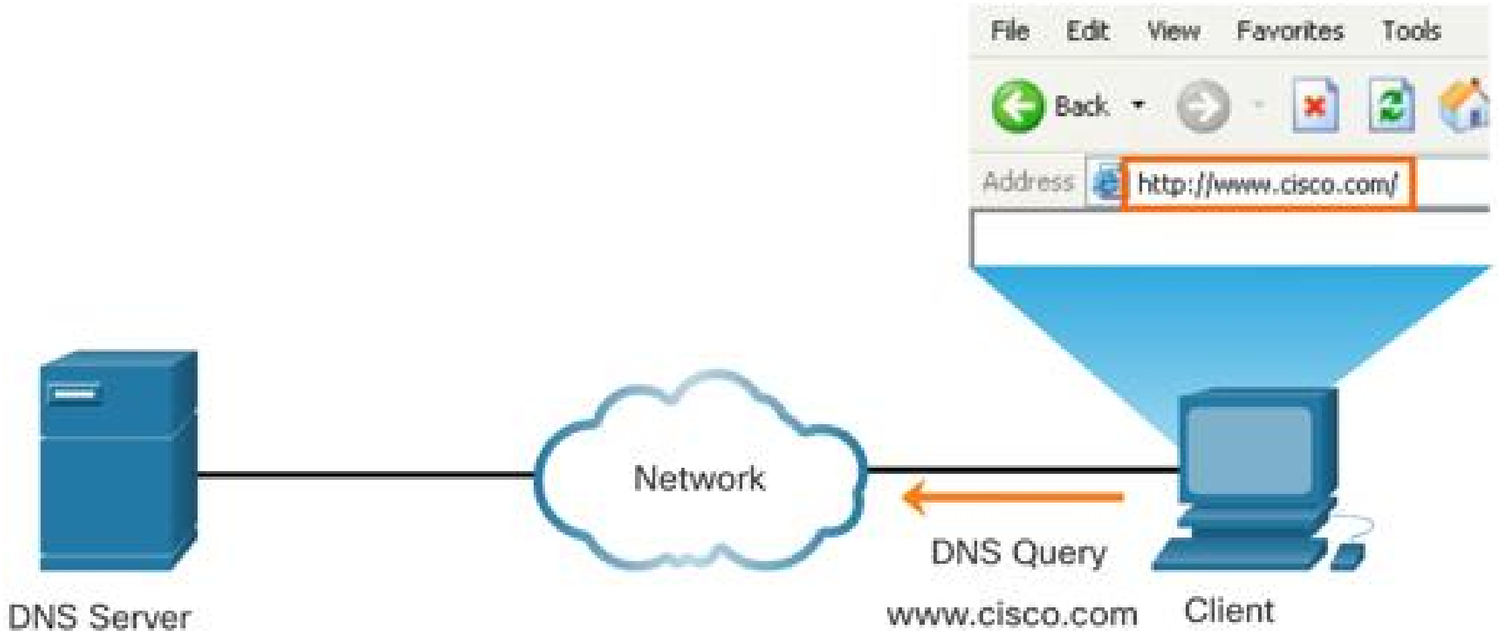
- Domain names were created to convert the numeric IP addresses into a simple, recognizable name.
- Fully-qualified domain names (FQDNs), such as http://www.cisco.com, are much easier for people to remember than 198.133.219.25.
- The DNS protocol defines an automated service that matches resource names with the required numeric network address. It includes the format for queries, responses, and data.

IP Addressing Services

Domain Name Service

Step 1

The user types an FQDN into a browser application Address field.

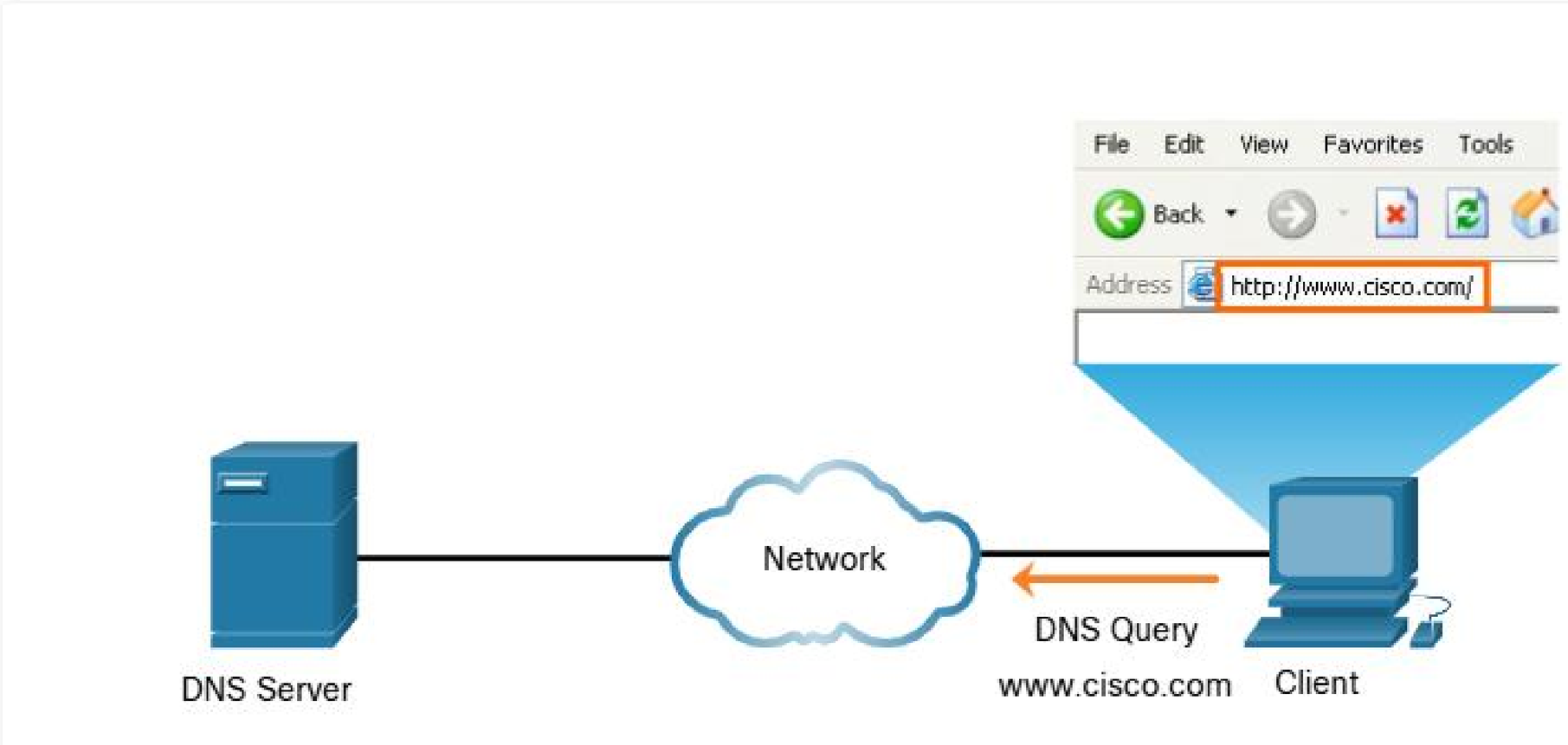


IP Addressing Services

Domain Name Service

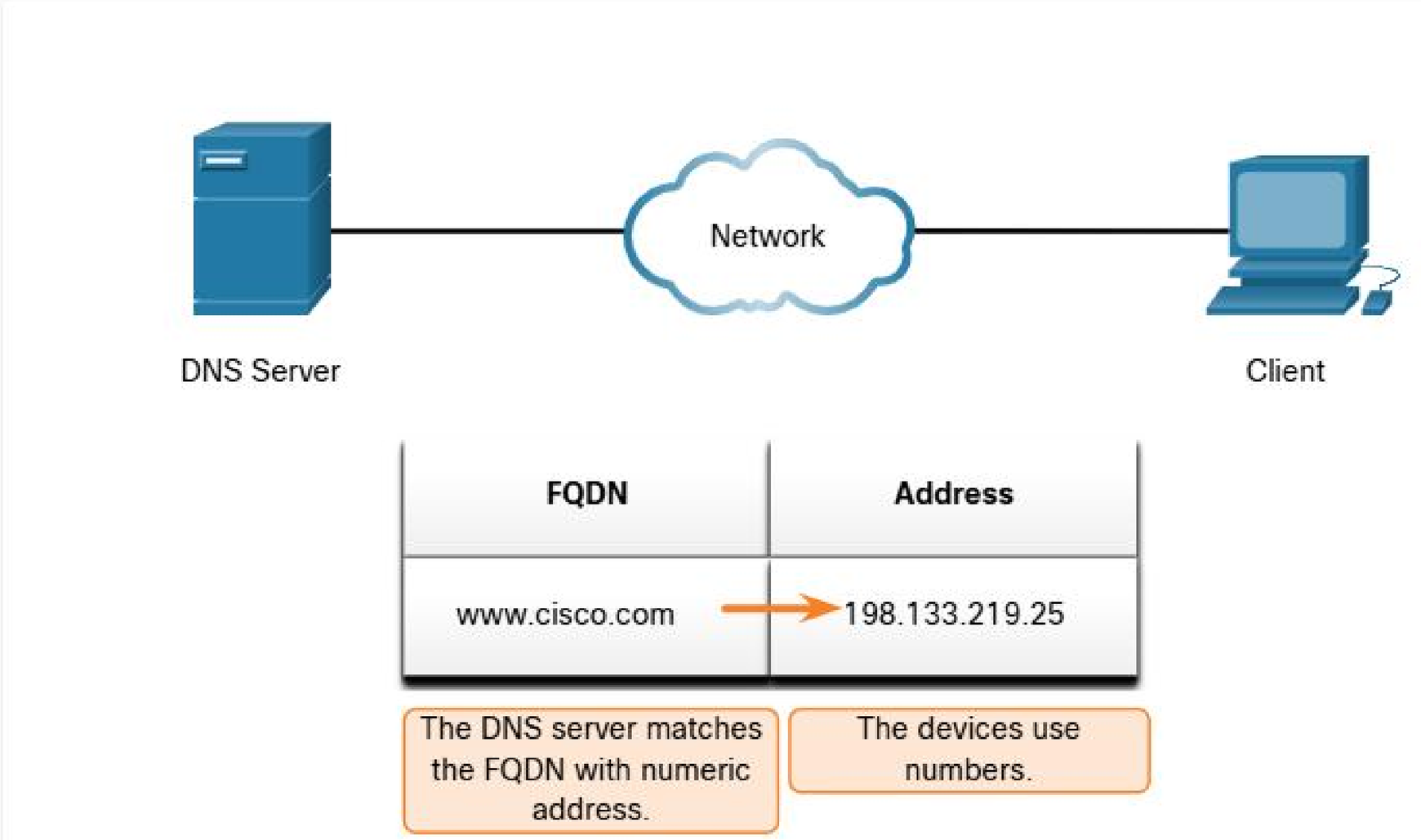
Step 2

A DNS query is sent to the designated DNS server for the client computer.



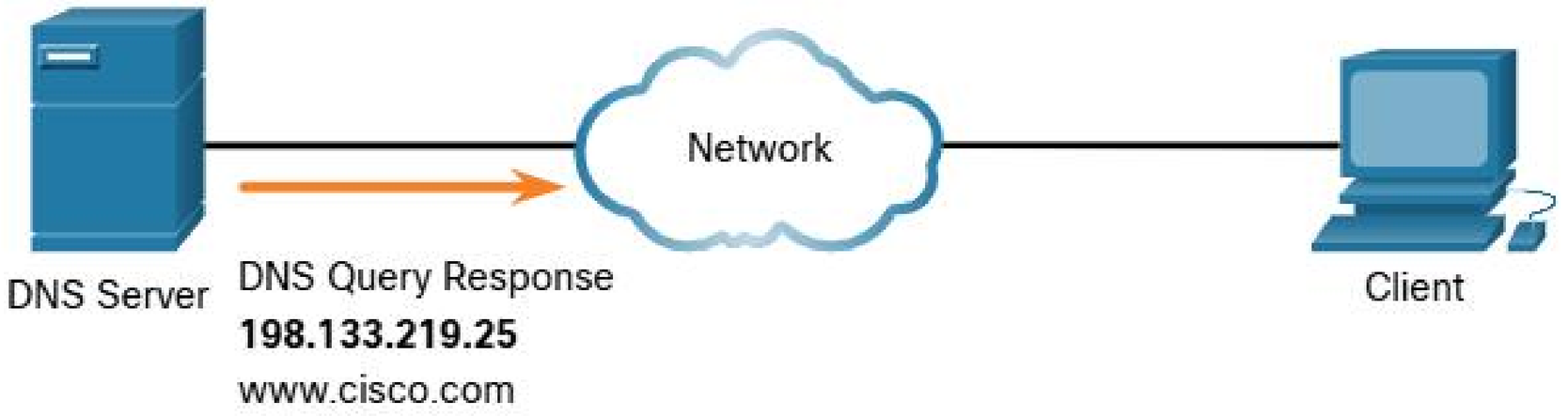
Step 3

The DNS server matches the FQDN with its IP address.



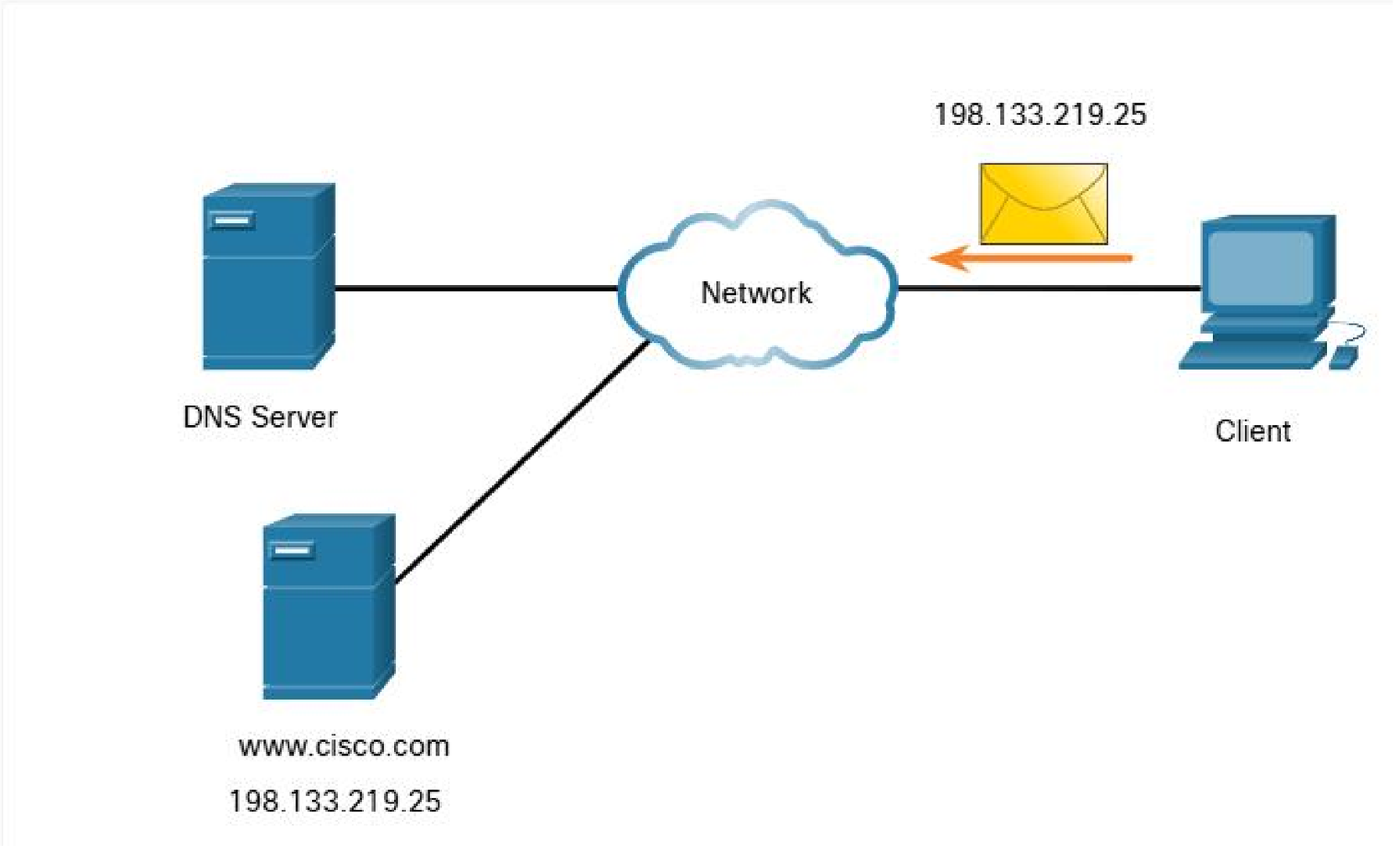
Step 4

The DNS query response is sent back to the client with the IP address for the FQDN.



Step 5

The client computer uses the IP address to make requests of the server.



DNS Message Format

The DNS server stores different types of resource records that are used to resolve names. These records contain the name, address, and type of record.

Some of these record types are as follows:

- **A** - An end device IPv4 address
- **NS** - An authoritative name server
- **AAAA** - An end device IPv6 address
- **MX** - A mail exchange record

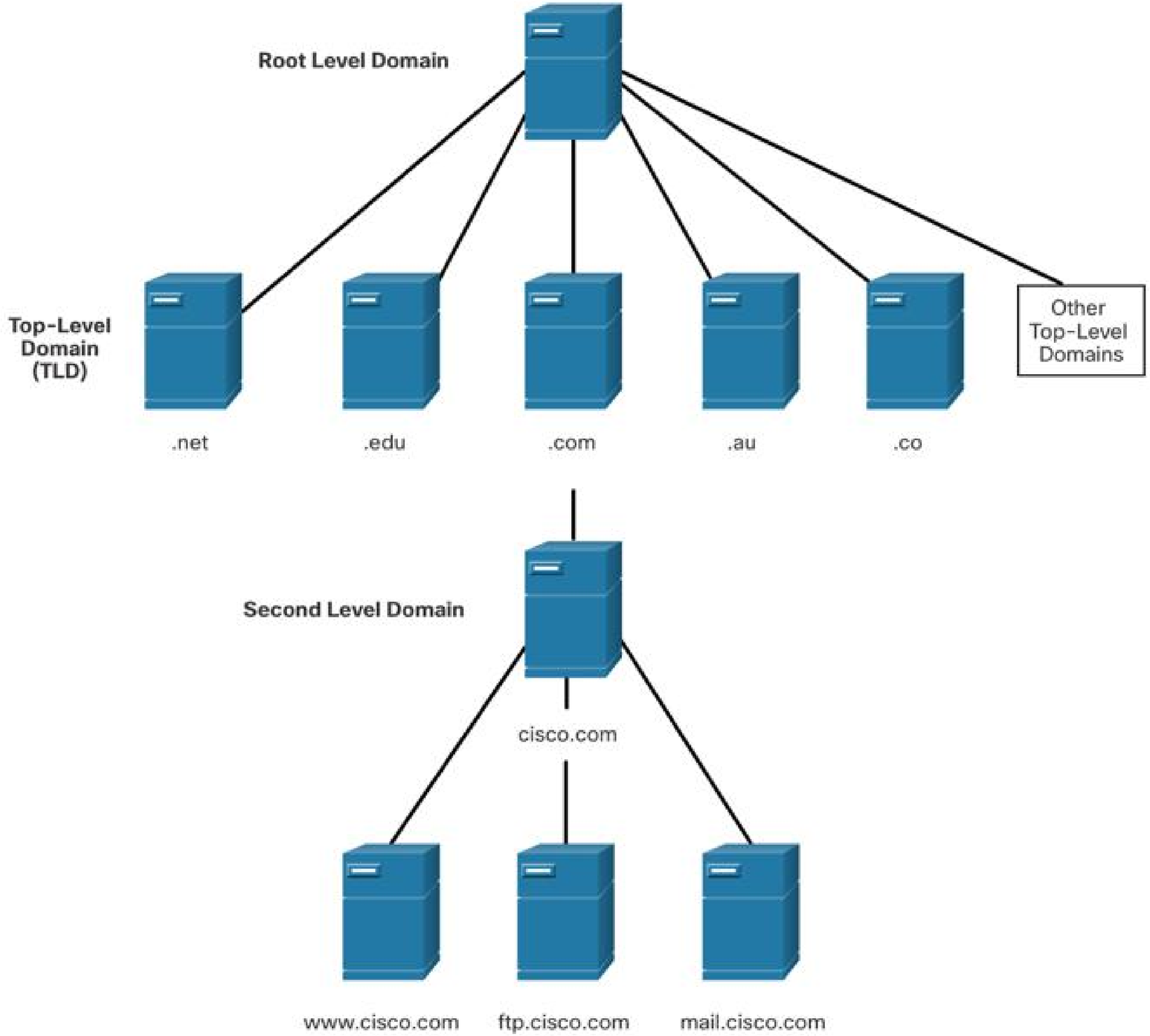
When a client makes a query, the server DNS process first looks at its own records to resolve the name. If it is unable to resolve the name by using its stored records, it contacts other servers to resolve the name.

After a match is found and returned to the original requesting server, the server temporarily stores the numbered address in the event that the same name is requested again.

DNS Message Format (Cont.)

DNS uses the same message format between servers, consisting of a question, answer, authority, and additional information for all types of client queries and server responses, error messages, and transfer of resource record information.

DNS message section	Description
Question	The question for the name server
Answer	Resource Records answering the question
Authority	Resource Records pointing toward an authority
Additional	Resource Records holding additional information


Root Level Domain
Top-Level Domain (TLD)
.net
.edu
.com
.au
.co
Other Top-Level Domains
Second Level Domain
cisco.com
www.cisco.com
ftp.cisco.com
mail.cisco.com

IP Addressing Services

The nslookup Command

- Nslookup is a computer operating system utility that allows a user to manually query the DNS servers configured on the device to resolve a given host name.
- This utility can also be used to troubleshoot name resolution issues and to verify the current status of the name servers.
- When the **nslookup** command is issued, the default DNS server configured for your host is displayed.
- The name of a host or domain can be entered at the **nslookup** prompt.

```
C:\Users> nslookup
Default Server:  dns-sj.cisco.com
Address:  171.70.168.183
> www.cisco.com
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    origin-www.cisco.com
Addresses:  2001:420:1101:1::a
          173.37.145.84
Aliases:  www.cisco.com
> cisco.netacad.net
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    cisco.netacad.net
Address:  72.163.6.223
>
```

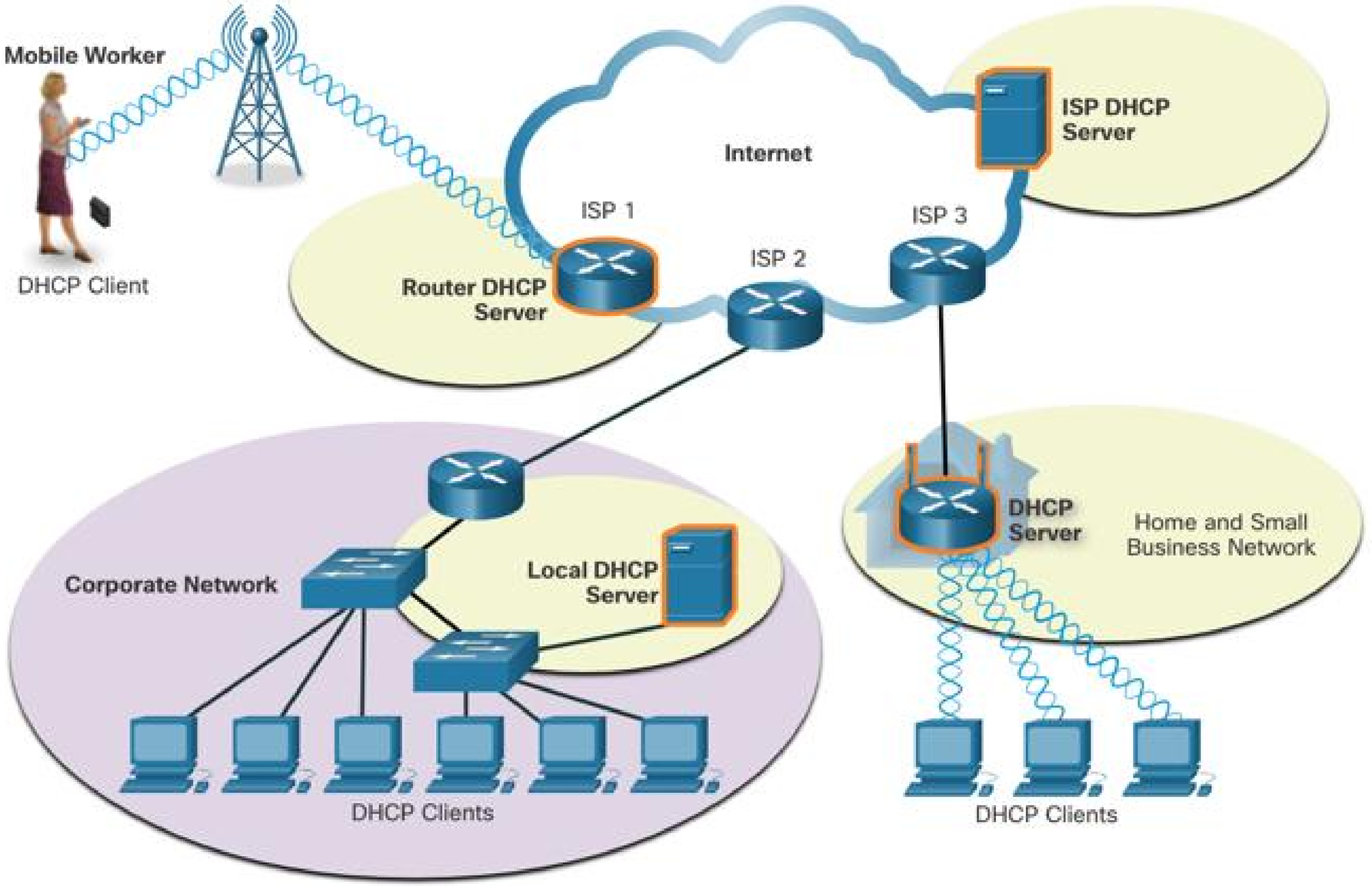
IP Addressing Services

Dynamic Host Configuration Protocol

- The Dynamic Host Configuration Protocol (DHCP) for IPv4 service automates the assignment of IPv4 addresses, subnet masks, gateways, and other IPv4 networking parameters.
- DHCP is considered dynamic addressing compared to static addressing. Static addressing is manually entering IP address information.
- When a host connects to the network, the DHCP server is contacted, and an address is requested. The DHCP server chooses an address from a configured range of addresses called a pool and assigns (leases) it to the host.
- Many networks use both DHCP and static addressing. DHCP is used for general purpose hosts, such as end user devices. Static addressing is used for network devices, such as gateway routers, switches, servers, and printers.

IP Addressing Services

Dynamic Host Configuration Protocol

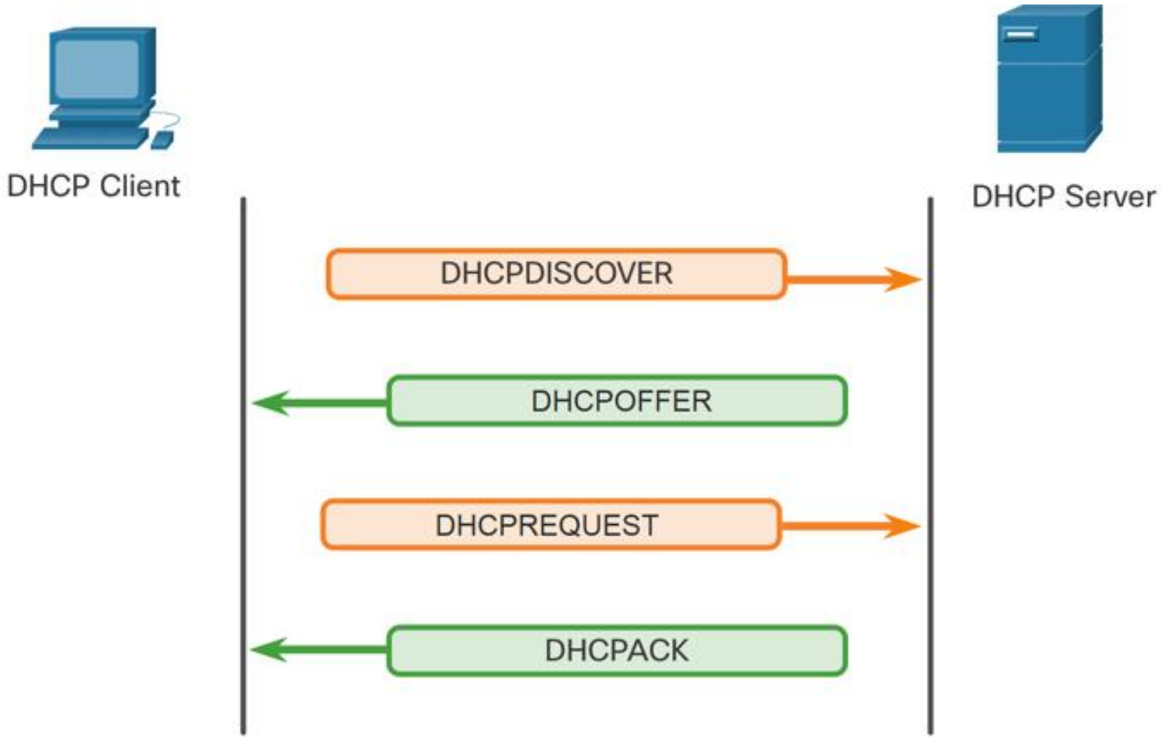


IP Addressing Services

DHCP Operation

The DHCP Process:

- When an IPv4, DHCP-configured device boots up or connects to the network, the client broadcasts a DHCP discover (DHCPDISCOVER) message to identify any available DHCP servers on the network.
- A DHCP server replies with a DHCP offer (DHCPOFFER) message, which offers a lease to the client. (If a client receives more than one offer due to multiple DHCP servers on the network, it must choose one.)
- The client sends a DHCP request (DHCPREQUEST) message that identifies the explicit server and lease offer that the client is accepting.
- The server then returns a DHCP acknowledgment (DHCPACK) message that acknowledges to the client that the lease has been finalized.
- If the offer is no longer valid, then the selected server responds with a DHCP negative acknowledgment (DHCPNAK) message and the process must begin with a new DHCPDISCOVER message.



When a DHCP server is configured, which two IP addresses should never be assignable to hosts? (Choose two)

A. network or subnetwork IP address
B. broadcast address on the network
C. IP address leased to the LAN
D. IP address used by the interfaces
E. manually assigned address to the clients
F. designated IP address to the DHCP server

Which two tasks does the Dynamic Host Configuration Protocol perform? (Choose two)

A. Set the IP gateway to be used by the network.
B. Perform host discovery used DHCPDISCOVER message.
C. Configure IP address parameters from DHCP server to a host.
D. Provide an easy management of layer 3 devices.
E. Monitor IP performance using the DHCP server.
F. Assign and renew IP address from the default pool.

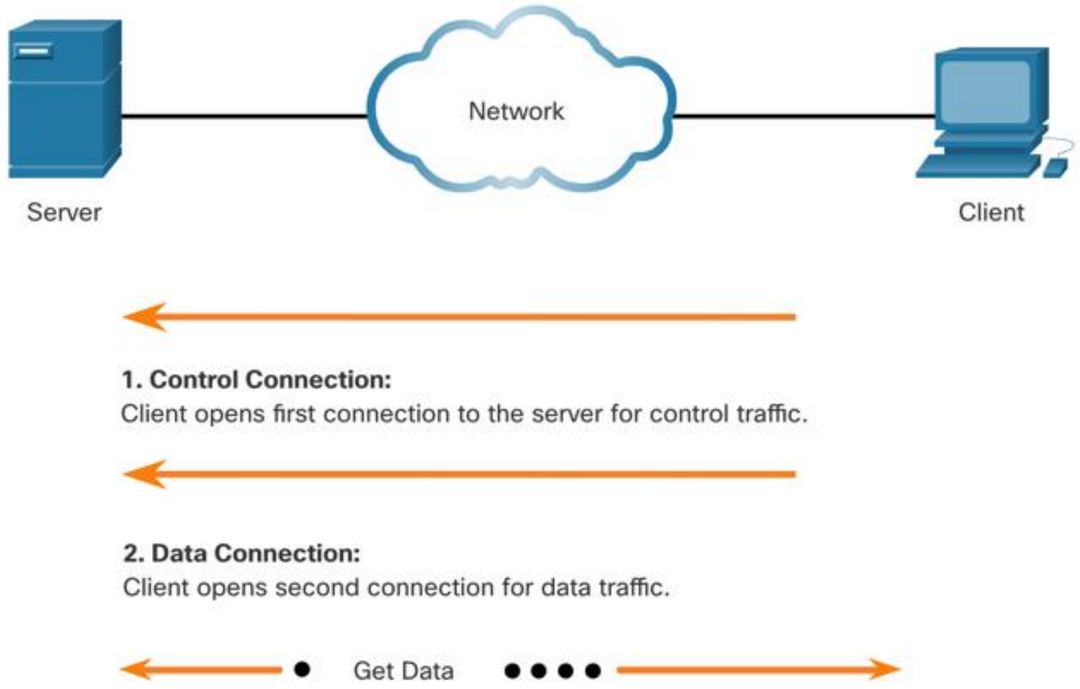
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File Sharing Services

File Sharing Services

File Transfer Protocol

FTP was developed to allow for data transfers between a client and a server. An FTP client is an application which runs on a computer that is being used to push and pull data from an FTP server.



Step 1 - The client establishes the first connection to the server for control traffic using TCP port 21. The traffic consists of client commands and server replies.

Step 2 - The client establishes the second connection to the server for the actual data transfer using TCP port 20. This connection is created every time there is data to be transferred.

Step 3 - The data transfer can happen in either direction. The client can download (pull) data from the server, or the client can upload (push) data to the server.

File Sharing Services

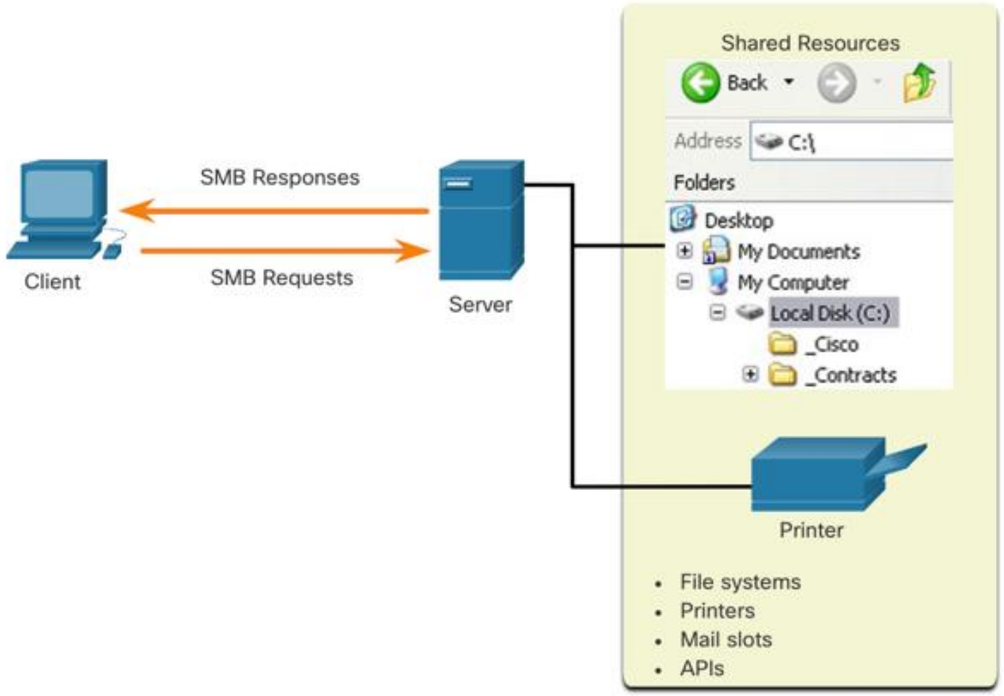
Server Message Block

The Server Message Block (SMB) is a client/server, request-response file sharing protocol. Servers can make their own resources available to clients on the network.

Three functions of SMB messages:

- Start, authenticate, and terminate sessions
- Control file and printer access
- Allow an application to send or receive messages to or from another device

Unlike the file sharing supported by FTP, clients establish a long-term connection to servers. After the connection is established, the user of the client can access the resources on the server as though the resource is local to the client host.



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THANK YOU!